

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1516

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Duration :60 Minutes

Code of Conduct

I Y SRAVAN KUMAR (192472289) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SRAVAN Y

Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

Problem:

A retail company needs real-time inventory tracking, demand forecasting, and personalized customer recommendations.

Solution:

- Collect sales, inventory, website, and customer data.
- Use **Apache Kafka** for real-time data streaming.
- Store large datasets using **Hadoop HDFS** or cloud storage.
- Use **Apache Spark** for large-scale data processing.
- Use **Spark Streaming** for real-time inventory updates.
- Use **Time-Series/ARIMA or LSTM models** for demand forecasting.
- Use **Collaborative Filtering** for personalized recommendations.
- Use **Power BI/Tableau** for business dashboards.

Data**Flow:**

Stores + Website → Kafka → Data Lake → Spark → ML Models → Dashboard/Recommendations

Application:

Used by **Amazon, Walmart, supermarkets, and e-commerce platforms** for inventory and customer personalization.

2. **Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

Problem:

A social media platform needs to detect fake accounts and misinformation from billions of interactions.

Solution:

- Collect posts, comments, likes, shares, and account activity.
- Use **Apache Kafka** for real-time event streaming.
- Use **Apache Spark Streaming** or **Apache Flink** for real-time processing.
- Store historical data in a **Data Lake/Hadoop** environment.
- Use **Machine Learning** to identify suspicious behavior.
- Use **Random Forest/XGBoost** for fake-account classification.
- Use **NLP and Transformer models** to detect misinformation.
- Generate alerts when suspicious activity is detected.

Data**Flow:**

User Activity → Kafka → Stream Processing → ML/NLP Models → Fraud Alert

Application:

Used in **social networking platforms, online communities, and content-sharing applications**.

3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

Problem:

Government departments need to share census, economic, and geospatial data securely while maintaining interoperability and privacy.

Solution:

- Collect data from different government departments.
- Store structured and unstructured data in a **Government Data Lake**.
- Use **ETL/ELT tools** for data integration.

- Use standardized formats such as **JSON, XML, CSV, and GeoJSON**.
- Use APIs for secure data sharing between departments.
- Apply **data governance policies** for ownership, quality, and standards.
- Use **role-based access control (RBAC)**.
- Encrypt data during storage and transmission.
- Use **data masking and anonymization** for sensitive citizen information.
- Maintain audit logs for monitoring access.

Data

Flow:

Government Departments → Data Integration → Data Lake → Governance & Security → Shared Data Platform

Application:

Used for **census management, public services, economic planning, smart cities, and national policy planning**.

4. Banking Fraud Detection Problem

A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

Problem: A bank processes millions of transactions and must detect fraudulent transactions with very low latency.

Solution:

- Collect transaction data from ATMs, cards, mobile banking, and online banking.
- Use **Apache Kafka** for real-time transaction streaming.
- Use **Apache Flink/Spark Streaming** for real-time processing.
- Store historical transactions in a **Data Lake/Data Warehouse**.
- Use **Random Forest, XGBoost, and Neural Networks** for fraud classification.
- Use **Anomaly Detection** to identify unusual transaction patterns.
- Use historical customer behavior for model training.
- Generate an immediate alert when a transaction is classified as suspicious.
- Continuously update the ML model using new transaction data.

Data Flow:

Transaction → Kafka → Stream Processing → Fraud ML Model → Risk Score → Alert/Block

Application:

Used in **banks, credit-card companies, digital wallets, UPI, and online payment systems**.

5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated

from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Problem:

Farmers need to analyze soil, weather, crop, drone, and sensor data to improve yield and reduce resource usage.

Solution:

- Collect data from **IoT sensors, drones, satellites, weather stations, and farm equipment**.
- Use **MQTT/Kafka** for collecting real-time sensor data.
- Store large datasets in a **Cloud Data Lake**.
- Use **Apache Spark** for large-scale data processing.
- Use **Machine Learning** for crop yield prediction.
- Use **Regression models** for yield forecasting.
- Use **Classification models** for crop disease detection.
- Use sensor analytics to determine irrigation and fertilizer requirements.
- Use dashboards to provide farmers with real-time recommendations.

Data Flow:

Sensors + Drones + Weather Data → Kafka/MQTT → Cloud Data Lake → Spark → ML Analytics → Farmer Dashboard

Application:

Used in **precision farming, crop monitoring, smart irrigation, disease detection, and yield optimization**.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of	Reasonable awareness	Limited awareness	Poor awareness

		security and operational risks			
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded: